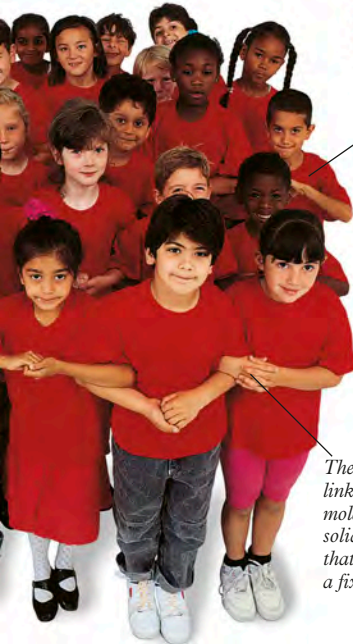


Making molecules

Atoms link up to make molecules. Two atoms of hydrogen join together to make one molecule of hydrogen gas.

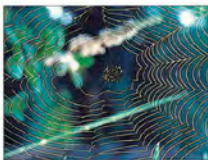


The molecules in a cool solid do not move around very fast. If a solid is heated, the molecules move around faster and faster.

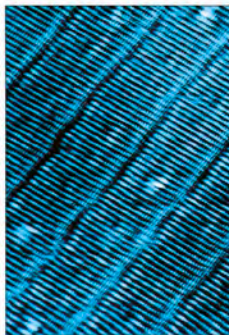
There are strong links between the molecules in a solid. This means that solids have a fixed shape.



A shell is hard and brittle.



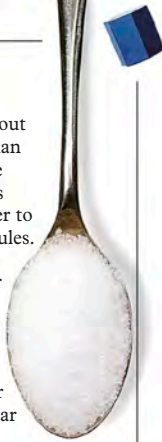
A spider's web is light and strong.



Larger than life

These long, thin wax molecules are shown about 3 million times larger than they are in real life. The tiny blue dots are atoms that have joined together to make up the wax molecules.

Sugar



Crystal clear

Grains of sugar are solid crystals. Their atoms are close together and arranged in a regular pattern called a lattice. Many solids that look smooth actually have a crystal structure, like these vitamin C tablets.



Vitamin C tablets



Vitamin C crystals, seen through a microscope.

Graphite and diamond

Graphite is soft and is used in pencils, and diamond is the hardest solid in the world. But diamond and graphite have a lot in common! They are two different forms of carbon. This means they have the same molecules, but they are arranged in different ways.

